

# K253 PRE-STAGE — Grace dual- $\rho$ NECESSARY derivation audit (extensive $\int$ additive $\int$ numerator; intensive $\int$ per-direction-power $\int$ denominator) at SUBSTRATE-MECHANISM-FORWARD tier

Keeper (Claude Opus 4.7)

2026-06-07 Sunday ~15:25 EDT (date-verified actual)

## K253 PRE-STAGE — Grace dual- $\rho$ NECESSARY derivation audit

### 0. Purpose

K-audit on Grace dual- $\rho$  NECESSARY derivation (grace\_dual\_rho\_extensive\_necessity\_derivation\_2026-06-07.md, Sun 2026-06-07 ~13:15 EDT actual) at SUBSTRATE-MECHANISM-FORWARD tier (existing standing per Cal #265 moratorium on category proliferation; no new sub-tier framing).

Grace's substantive substrate-architectural derivation closes a substantive substrate-mechanism gate for Casey-named principle #9 ("Substrate Floor / Scale-Not-Spectrum Reach"): proves the asymmetric dual- $\rho$  structure is substrate-architecturally NECESSARY rather than coincidental. This substantively grounds the substrate-floor scope-principle at substrate-mechanism level — substantive substrate-architectural promotion candidate.

### 1. F59 v0.2 substantive content (Grace)

#### 1.1. The substantive substrate-architectural derivation

Extensive half (PROVEN via T2488): 1. Extensive  $\int$  additive (definitional + standard physics) 2. T2488 cell-counts ADD: baryon = meson + 1 ( $n_C + Z_2$  bit); the +1 is added, the  $Z_2$  bit adds one cell 3. T2488 additivity  $\int$  mass is extensive (derived, not assumed) 4. Extensive  $\int$  total integral  $\int$  volume-form over complex- $n_C = \pi^{n_C}$  5. Total enters multiplicatively, in the numerator 6.  $\int$  conformal- $\rho$  / volume side NECESSARILY carries  $\pi^{n_C}$  in numerator and ADDS mass

Intensive half (by definition + form-confirmed): 1. Casimir = eigenvalue of operator on single irreducible K-type 2. Eigenvalue = property of mode, independent of system size (per-mode density) 3. Per-mode density = intensive by definition 4. Spectral observable = per-direction density / normalization, enters as ratio in denominator 5. Form confirms:  $m_\mu/m_e = (24/\pi^2)^{C_2}$  is per-direction factor  $(24/\pi^2)$  raised to direction-count  $C_2$  — intensive fingerprint 6.  $\pi^2$  in denominator because per-mode normalization divides 7.  $\int$  compact- $\rho$  / spectral side NECESSARILY carries  $\pi$  in denominator and normalizes/MEASURES

Asymmetry NECESSARY: Extensive (total) and intensive (per-unit) are categorically distinct, so opposite roles by necessity, not mirror, not coincidence. The structural fingerprint is additive (extensive) vs per-direction-power (intensive) in the substrate-mechanism forms themselves.

#### 1.2. Substantive substrate-architectural grounding for Casey-named principle #9

Casey-named principle #9 ("Substrate Floor / Scale-Not-Spectrum Reach") states substrate reaches SCALES via ground-state mass = bulk volume +  $Z_2$  bit (kinematics); does NOT reach SPECTRA (excited states) or RATES (decays).

Grace's substantive substrate-mechanism FORWARD derivation grounds this at substrate-mechanism level: - Mass = extensive measure (total) = scale-setting (kinematics) - Spectrum + decay rates = intensive (per-mode/per-channel) = substrate normalizes but does NOT generate - The reach-bound IS the extensive/intensive dichotomy, derived from T2488 additivity + Casimir definition

This substantively promotes Casey-named principle #9 from SCOPE-PRINCIPLE (parallel to Five-Absence Predictions Set) to potential SUBSTRATE-MECHANISM-FORWARD principle (parallel to Casey #7 D\_IV<sup>5</sup> Rigidity or Casey #14 Substrate-Predicted 3+1 Minkowski Signature) with substantive substrate-architectural derivation behind it.

Cal call on promotion path: scope-principle vs substrate-mechanism FORWARD principle. The derivation substantively exists; the classification is Cal's audit-discipline call.

## 2. F1 — Mathematical content soundness

### 2.1. T2488 additivity input

T2488 (Lyra\_F57\_\* + Keeper K249 + Elie Toy 4022 + Grace order-2): cells = bulk  $n_C$  +  $Z_2$  bit. Cell-count is built by adding bulk wrap ( $n_C$ ) and spinor twist (1). The +1 is additive in the operator-language form. DERIVED Sunday (four-way convergence).

T2488 additivity is the substantive input Grace uses; the additivity is not assumed but derived content of T2488. F1 PASS on input verification.

### 2.2. Extensive $\int$ additive

Standard thermodynamics + statistical mechanics definition. Volume + charge + energy add; temperature + density don't. T2488's cell-count additivity puts mass in the extensive class. F1 PASS on definitional bridge.

### 2.3. $\int$ volume-form over complex- $n_C = \pi^{n_C}$

Standard complex-analytic measure-theory result:  $2n_C$ -real-dim ball or Gaussian volume integral with substrate-natural normalization closes to  $\pi^{n_C}$  (matches T2487 three-route derivation). F1 PASS on measure-theory standard.

### 2.4. Casimir = eigenvalue per K-type

Standard Lie-algebra representation theory: Casimir operator  $C$  acts as scalar  $\lambda_C$  on irreducible representation  $V_\lambda$  via Schur's lemma. Eigenvalue  $\lambda_C$  is per-K-type (intensive). F1 PASS on representation-theory standard.

### 2.5. $(24/\pi^2)^{C_2}$ per-direction-power fingerprint

T190  $(24/\pi^2)^{C_2} = m_\mu/m_e$  at Tier 1 EXACT  $\sim 10^{-5}$  precision. The form is per-direction factor  $(24/\pi^2)$  raised to direction-count  $C_2 = 6$  — intensive fingerprint per Grace's definition (per-mode density accumulated as power, not sum). F1 PASS on form-verification.

F1 verdict: substantive substrate-architectural derivation soundness PASSES. All inputs DERIVED or standard. Conclusion follows substantively from inputs.

## 3. F2 — K231c “derived not relabeled” check

Grace derives the asymmetry NECESSITY from substantive ingredients (T2488 additivity + Casimir definition + standard measure theory + representation theory). NOT a relabel of: - F59 v0.1 (which articulated the asymmetric reading but left necessity OPEN) - T2487 (which establishes mass=volume but doesn't derive the asymmetry) - F61 (which states the reach-bound but doesn't derive its substrate-mechanism)

F59 v0.2 substantively DERIVES the substrate-mechanism that F61 only states. Forward derivation, not relabel. F2 PASS.

#### 4. F3 — Substantive substrate-architectural distinctness

Grace's derivation is substantively distinct from: - F61 (states reach-statement; doesn't derive substrate-mechanism) - F59 v0.1 (articulates asymmetry; leaves necessity open) - T2487 (mass = volume substantive identity) - T2488 (cell-count via  $Z_2$  bit empirical anchors) - Elie a\_k  $\rightarrow$  induced gravity map (Sunday 2026-06-07 ~14:50 EDT) — different substantive substrate-architectural content; complementary

This is substantive new substrate-mechanism FORWARD content: the substrate-architectural NECESSITY of the dual- $\rho$  asymmetry that grounds Casey-named principle #9.

F3 PASS on distinctness.

#### 5. F4 — Audit-category disposition (per Cal #265 moratorium)

SUBSTRATE-MECHANISM-FORWARD tier (existing standing per Cal stack). No new audit-category proposed (Cal #265 moratorium absorbed).

Distinguished from: - SUBSTRATE-MECHANISM-FORWARD CANDIDATE-LEAD (K231c, ratified): Grace's content is DERIVED, not candidate-lead awaiting derivation - SUBSTRATE-MECHANISM-FORWARD CONVENTION-PIN (K231a, 9th category ratified): Grace's content is mechanism derivation, not convention identification - SUBSTRATE-MECHANISM-FORWARD CLOSURE (K252 framing retracted): Grace's content is FORWARD, not closure of previously-open item

K253 PRE-STAGE at SUBSTRATE-MECHANISM-FORWARD tier per Cal stack standing. No new sub-tier framing. F4 PASS per moratorium.

#### 6. K253 verdict + falsifier carried

K253 CONDITIONAL PASS at SUBSTRATE-MECHANISM-FORWARD tier (existing standing).

Sub-verdicts: - F1 (mathematical content): PASS — all inputs DERIVED or standard; conclusion follows substantively - F2 (K231c “derived not relabeled”): PASS — substantive forward derivation, not relabel - F3 (distinctness): PASS — substantively new substrate-mechanism FORWARD content - F4 (audit-category per Cal #265): PASS — existing SUBSTRATE-MECHANISM-FORWARD tier; no new category proposed

Falsifier carried (per Grace F59 v0.2 Section 4): 1. If mass were intensive (per-mode ratio, not additive) — T2488 already DERIVES additivity; falsifier not triggered 2. If Casimir were extensive (total, not per-mode eigenvalue) — definitional; falsifier not triggered 3. Genuine test: counterexample — an additive (extensive) observable riding a normalization denominator, OR a per-direction-power (intensive) observable adding. T2483 dual- $\rho$  classification is the place to look for one.

If counterexample is found, the substrate-architectural NECESSITY breaks; F59 v0.2 fails; Casey-named principle #9 substantive substrate-mechanism grounding falls back to scope-principle level.

#### 7. Substantive substrate-architectural promotion-path implication

Casey-named principle #9 substantive substrate-architectural status: - Pre-K253: SCOPE-PRINCIPLE (parallel to Five-Absence Predictions Set) — what substrate does NOT reach - Post-K253 (if Cal cleared): SUBSTRATE-MECHANISM-FORWARD principle (parallel to Casey #7 D\_IV<sup>5</sup> Rigidity / Casey #14 Substrate-Predicted 3+1 Minkowski) — what substrate substantively DOES (sets extensive ground-state mass scale via bulk volume +  $Z_2$  bit) AND substantively does NOT (intensive per-mode spectrum + rates), DERIVED from substrate-mechanism extensive/intensive dichotomy

The promotion-path requires: - Cal cold-read PASS on K253 + F59 v0.2 substantive content - Casey-named principle #9 classification update (scope-principle vs substrate-mechanism FORWARD principle) - Strong-Uniqueness Theorem absorption (Lyra call): substrate-mechanism FORWARD content substantively strengthens substrate-architectural uniqueness argument

Substantive substrate-architectural significance: if promotion clears, Casey-named principle #9 substantively strengthens the substrate-architectural unique-forcing argument by adding extensive/intensive dichotomy as substrate-mechanism FORWARD constraint on  $D_{IV}^5$ .

## 8. Cross-link map

K253 cross-links: - T2487 + T2488 DERIVED weekend advances (substantive substrate-architectural inputs) - F59 v0.1 (asymmetric dual- $\rho$  articulation)  $\rightarrow$  F59 v0.2 (NECESSARY derivation) - F61 / Casey-named principle #9 substrate-floor scope-principle (now substantively grounded) - T190 ( $m_\mu/m_e = (24/\pi^2)^{\{C_2\}}$  per-direction-power fingerprint) - Elie  $a_k \rightarrow$  induced gravity map (Sunday): complementary substantive substrate-mechanism content; substrate-natural Sakharov induced gravity coefficients  $a_0 = 225 (\Lambda) + a_1 = -1875$  (Einstein-Hilbert) - Vol 16 Ch 9 absorption candidate: extensive/intensive dichotomy in mass mechanism chapter - Vol 16 Ch 13 absorption candidate: substrate-floor as new C-criterion of Strong-Uniqueness Theorem - Cal #265 STANDING (no new audit-category proliferation): K253 at existing SUBSTRATE-MECHANISM-FORWARD tier

## 9. Open items + Cal cold-read gate

Open for Cal cold-read: - K253 PRE-STAGE verdict (CONDITIONAL PASS at SUBSTRATE-MECHANISM-FORWARD tier) - Casey-named principle #9 promotion-path classification call (scope-principle vs substrate-mechanism FORWARD principle) - F59 v0.2 substantive content cold-read verification - Falsifier survey (T2483 dual- $\rho$  classification counterexample search)

Open for Keeper lane: - Vol 16 Ch 9 absorption support (Lyra primary): extensive/intensive dichotomy substantive substrate-architectural content - Vol 16 Ch 13 Strong-Uniqueness Theorem absorption: substrate-floor as new C-criterion CANDIDATE - Cross-link to Elie  $a_k \rightarrow$  induced gravity map (different substantive substrate-mechanism content; complementary)

Open for Lyra lane: - F60 Step 2 substantive gravity climb (concrete handle from Elie  $a_k$  map) - Strong-Uniqueness Theorem absorption call (substrate-floor as new C-criterion)

Open for Grace lane: - Field-theory  $\rho$ -classification sweep across SM sectors (extensive/intensive) - Mass-Gap setup grounded on extensive floor (Millennium-adjacent substantive substrate-architectural thread)

## 10. Closure

K253 PRE-STAGE on Grace F59 v0.2 NECESSARY derivation at SUBSTRATE-MECHANISM-FORWARD tier (existing standing per Cal #265). CONDITIONAL PASS pending Cal cold-read.

Grace substantively derives the asymmetric dual- $\rho$  structure as NECESSARY (extensive  $\frac{1}{2}$  additive  $\frac{1}{2}$  numerator-ADDS; intensive  $\frac{1}{2}$  per-direction-power  $\frac{1}{2}$  denominator-normalizes) from T2488 additivity + Casimir definition + standard measure theory + representation theory. Substantively grounds Casey-named principle #9 substrate-floor scope-principle at substrate-mechanism level. Substantive substrate-architectural promotion-path candidate for Casey-named principle #9 from scope-principle to substrate-mechanism FORWARD principle.

Falsifier explicit: counterexample (additive observable in denominator OR per-direction-power adding) would break the substrate-architectural NECESSITY.

— Keeper, Sun 2026-06-07 ~15:25 EDT (date-verified actual)